

## Product Specification Sheet

Ingredient: Vitamin B12

Intended Use: Food Supplement Applications

Product Code: 64

Revision Date: 06/14/2025

Country of Origin: Brazil

### 1. Product Description

Vitamin B12 is commonly used in the formulation of food supplements due to its functional and nutritional properties.

### 2. Composition

- Active Ingredient: Vitamin B12
- Purity:  $\geq 95\%$  (unless otherwise specified)

### 3. Physical & Organoleptic Characteristics

- Appearance: Crystalline powder
- Color: Red or dark pink
- Odor/Taste: Characteristic or bland
- Solubility: Soluble or dispersible in water
- Bulk Density: 0.35-0.65 g/cm<sup>3</sup>
- pH (1% solution @ 25C): 4.0-7.5 (typical range)

### 4. Phytochemical Parameters (if applicable)

- Identification: Positive

- Assay: Conforms to standard
- Heavy Metals: <10 ppm
- Lead (Pb): <3 ppm
- Arsenic (As): <1 ppm
- Mercury (Hg): <0.1 ppm

## 5. Microbiological Specification

- Total Plate Count: ≤ 10,000 CFU/g
- Coliforms: < 10 CFU/g
- Salmonella: Absent / 25g
- Yeasts & Molds: ≤ 100 CFU/g

## 6. Packaging

- Food-grade, moisture-resistant packaging
- Net weight: 20 kg or 25 kg
- Customized packaging available

## 7. Shelf Life and Storage

- Shelf Life: 24 months from production date
- Storage Conditions: Store in a cool, dry environment (<25C, RH <65%), away from sunlight and strong odors. Keep sealed.

## 8. Regulatory Status

- Complies with international food safety standards (EU/FDA/Codex)
- Allergen: May contain or be processed in facilities handling common allergens